Maintenance for Laser Powder Bed Fusion 331

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Class Outline

Laser Powder Bed Fusion
Software Access Levels
Machine Components
Maintenance Approaches
Maintenance System Review
Software System Maintenance
Weekly Maintenance
Daily Maintenance
As-Needed Maintenance
Other Maintenance Considerations
L-PBF Maintenance Review

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Laser Powder Bed Fusion

In laser powder bed fusion (L-PBF), an additive manufacturing (AM) machine directs a laser beam to heat and fuse powdered material to build a part layer-by-layer. In order to optimize quality and repeatability, an L-PBF operator must follow basic maintenance activities necessary for the care and repair of L-PBF machines.

The manufacturing objectives of an organization will frequently dictate their maintenance approach, or planned maintenance optimization (PMO) strategy. By doing maintenance on equipment while it is still in good operating condition, preventative maintenance strategies like inspection and cleaning can reduce breakdowns, which can help manufacturers who operate on rigid production schedules. Planned maintenance (PM), a type of preventative maintenance, includes checking alarms and alert logs and replacing worn parts on a scheduled or as-needed basis. Finally, recording the results of any maintenance activity can aid in predictive maintenance (Pdm) approaches, which require gathering data on an L-PBF machine's operation to service a machine before problems develop.

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Figure 1: L-PBF machine operation, courtesy of GE Additive.

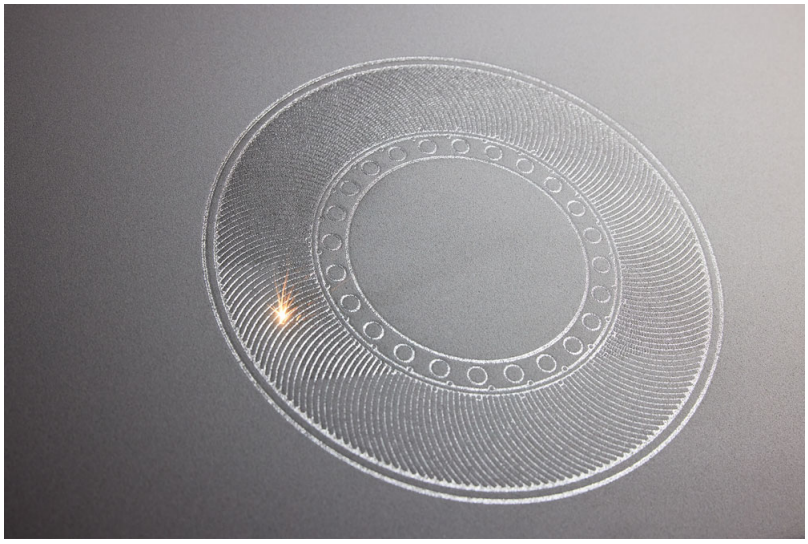

Figure 2: L-PBF machine operation, courtesy of GE Additive.

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Software Access Levels

Typically, an L-PBF operator will access the build controls and process controls through the machine's main computer. Different personnel may have different software access levels depending on the task. Most operators will have access to basic software that will enable them to carry out common pre-build maintenance tasks, before the build has started, and post-build maintenance tasks that may need to occur afterwards.

Other access levels can include a technician level for more advanced repairs and an administrator level that gives access to all machine functions, including tasks like system testing, calibration, material changing, and configuration. However, more advanced software access can increase the chances of unapproved software changes that might damage the machine, void the warranty, or create a safety hazard around hazardous substances. For these reasons, operators should not attempt to access L-PBF machine functions without proper training.

Figure 1: Many common maintenance tasks are performed with operator login access.

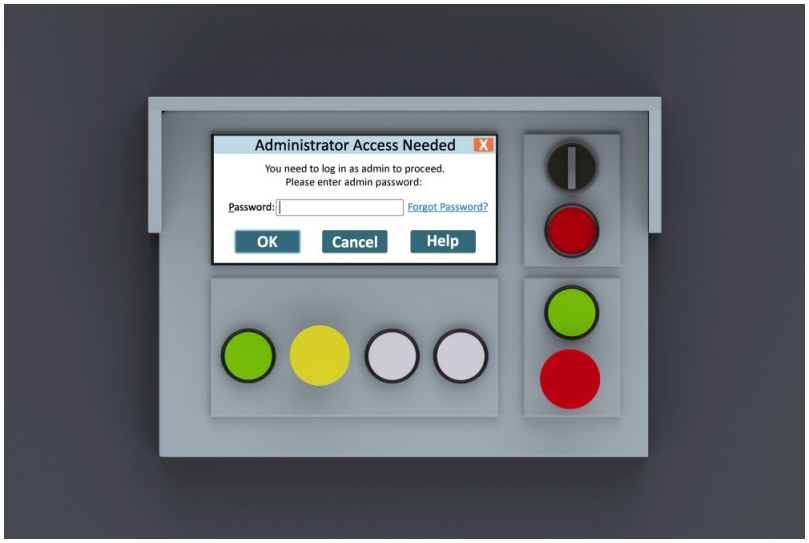

Figure 2: More advanced maintenance issues may need administrator access.

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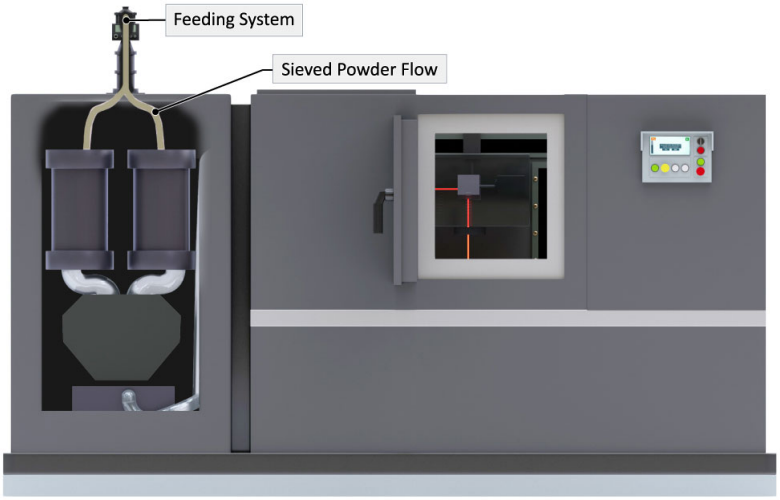
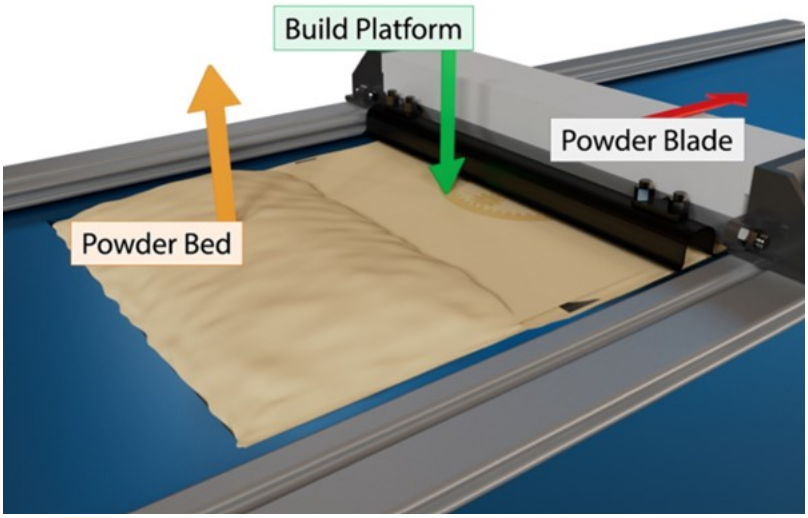
Machine Components



To begin the L-PBF process, an operator must transfer sieved powder into the powder hopper through the feeding system. Since most metal powders pose a respiratory and fire hazard, an operator should examine the safety data sheet (SDS) before attempting to load the powder. For example, cleaning spilled metal dust with compressed air creates an inhalation risk and could create an explosive dust cloud.



The main purpose of the powder hopper is to dispense, or dose, the build platform with a specific amount of powder. After each dosing from the hopper, a powder blade, or recoater, spreads the fresh powder uniformly over the build platform if possible, the powder blade typically pushes excess powder into the overflow chambers to be recirculated.

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The laser uses a lens to focus the beam on the build surface. Typically, the lens, mirror system, and laser are located behind a protective window, which protects the entire optical system from contamination. Any dust residue or scratches in this protective window can affect laser focus and effectiveness.

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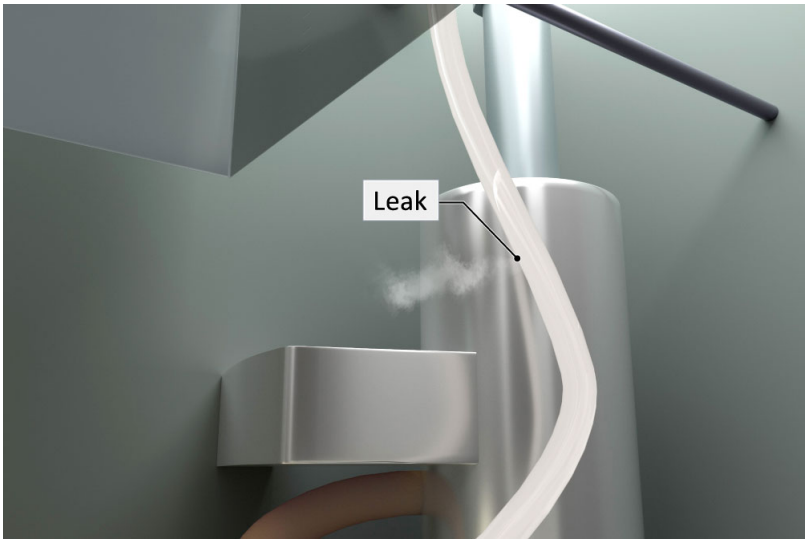
Maintenance Approaches

When carrying out any maintenance task, including cleaning the area around the L-PBF machine, operators should wear the proper personal protective equipment (PPE). This includes non-static, full-length clothing, a full-face respirator, eye protection, and protective gloves.



Inspection tasks involve examining machine components to ensure that they conform to expected standards. Tasks may include examining the machine for wear, leaks, distortion, and corrosion.

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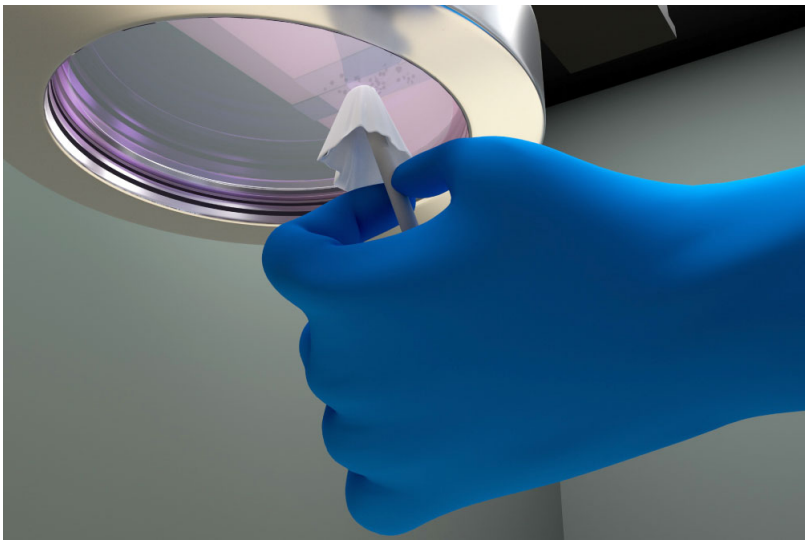
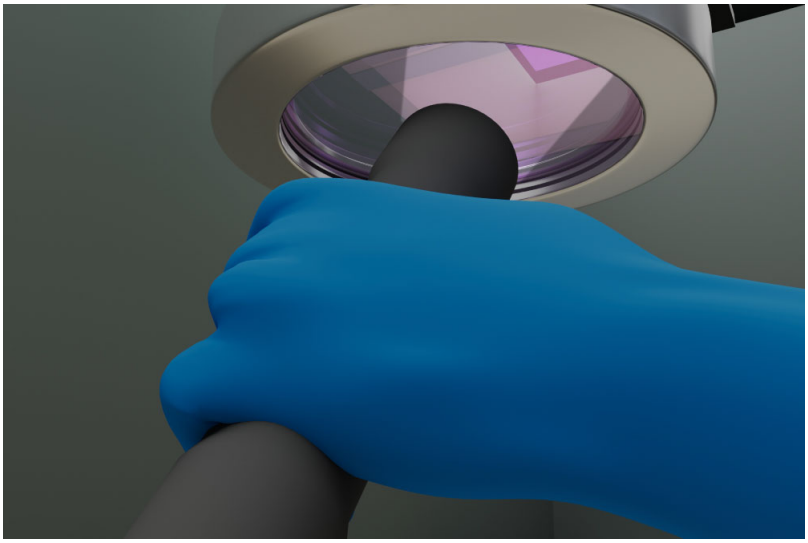


Cleaning tasks include the removal of dirt and impurities from the L-PBF machine.



Checking can include verifying that an L-PBF machine, part, or system meets manufacturing specifications for things like proper laser beam size and power.

Equipment	Maintenance Activity	Skill Level	Completed
Process room	Clean	Operator	
Build chamber viewing window	Inspect	Operator	
Top and bottom door seals	Check	Operator	
Safety signs and laser safety labels	Inspect	Operator	
User guide	Check	Operator	
Powder flasks & large safety change filters	Check	Operator	
Chiller settings	Check	Operator	
Vacuum pump oil level	Check	Operator	
Inert gas supply safety	Check	Operator	
Lens protection window	Clean	Operator	

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Recording specific information on machine performance or maintenance tasks is an important part of machine setup and maintenance. Records or logs allow any personnel to confirm that repairs, software configuration, and other tasks comply with manufacturing procedures and safety policies. Effective record keeping can also help to predict future repairs and potential upgrades.

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Lesson: 5/11

Maintenance System Review

1. An organization's manufacturing objectives frequently dictate their maintenance approaches. **TRUE**
2. Typically, an L-PBF operator will access the build and process data through the machine's build chamber. **FALSE**
3. An operator must transfer sieved powder into the powder hopper through the feeding system. **TRUE**
4. Record keeping can interfere with predicting future repairs. **FALSE**

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Software System Maintenance

Most L-PBF machine software connects to a separate computer or network. Some manufacturers may use a single wide area network (WAN), which connects devices in multiple facilities or locations, such as a general internet connection. A WAN connects L-PBF machines with hardware such as personal computers (PCs), routers, and onsite servers, as well as data collection software and offsite data storage repositories like the cloud. Other manufacturers may choose to isolate equipment to a local area network (LAN) within a single facility or area, which can still enable L-PBF machines to transfer production data internally while restricting internet access.

Connecting L-PBF machines to a network has both benefits and challenges. Any smart sensor on a machine collects and stores information to track and identify potential maintenance issues, which enables personnel to monitor a WAN-connected machine's status from any geographic location. The more accessible the data, however, the more vulnerable L-PBF machines are to cyber attacks that can disrupt or disable machines, or even gain illegal access to proprietary information. Part of a regular maintenance routine should include checking for any potential vulnerabilities and ensuring all cybersecurity software is up-to-date, which will increase the cyber resiliency of the L-PBF machine and network.

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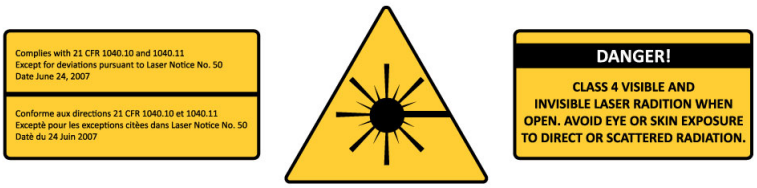
Weekly Maintenance



Weekly L-PBF maintenance tasks include inspecting the build chamber window for any cracks or damage while the L-PBF machine is powered down. Afterwards, an operator can check the door seals for any powder build up and remove it, if necessary, with a vacuum cleaner. Operators also regularly check the condition of safety labels to make sure they are clean and visible. Safety labels help L-PBF operators protect themselves from hazards associated with laser light radiation, gas, and powder exposure in addition to making sure the L-PBF machine complies with international standards.

Other weekly tasks include inspecting powder containers and hoppers for any damage or leaks. Make sure valves, clamps, and latches operate smoothly in both open and closed positions. In the closed position, check valves to make sure they seal properly with no leakage. An operator should report or replace any damaged or defective powder containers or associated equipment.

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Daily Maintenance

After a part build, when the L-PBF machine is powered down, an operator should clean the build chamber using a damp cloth with a mild, non-abrasive cleaning solution. Use special wipes to clean the touch screen and other areas of the control panel.

Courtesy of GE Additive

Daily inspection should include inspecting the exterior of the machine for any fluid leaks or noticeable damage, especially around the door and control panel. Also, check system hoses for signs of liquid or powder leaks, especially around the coolant system.

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When the machine is powered up but not building a part, operators should check important safety alerts to make sure they operate properly. For example, unlock the build chamber door and check for a door-open alert, and activate the emergency stop (E-stop) and check for the E-stop alert. Other important safety alerts to check include build chamber atmosphere levels since L-PBF machines use inert gases such as argon and nitrogen to control temperature and humidity to create an inert environment.

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As-Needed Maintenance

While many L-PBF machine maintenance tasks occur on a regular schedule, some only need to be done when the need arises. In addition to cleaning the build chamber surface after each use, it may need a deeper clean from time to time. When the machine is powered down, inspect the area behind the build chamber liner for any loose powder or condensation. Vacuum any loose powder. Be careful not to bump any mechanism or belts, which could affect printing accuracy. The frequency of deep cleanings depends on inspection findings over time. For example, if operators rarely find loose metal powder, frequent deep cleanings may not be necessary.

It is also important to check beneath L-PBF machines and in material storage and handling areas and, if necessary, clean them with an anti-static brush and dustpan or vacuum. For example, things like light fixtures, flooring, furniture, and air ducts can accumulate metal powder over time. With inert gas supply equipment, operators should check that the hoses and associated pipes remain free of damage and do not obstruct walkways.

Once the L-PBF machine is powered up, but before building, an operator should check the build chamber lights to make sure they shine correctly, and that the protective windows are clean.

Figure 1: Bumping any mechanism or belts while cleaning could affect printing accuracy.

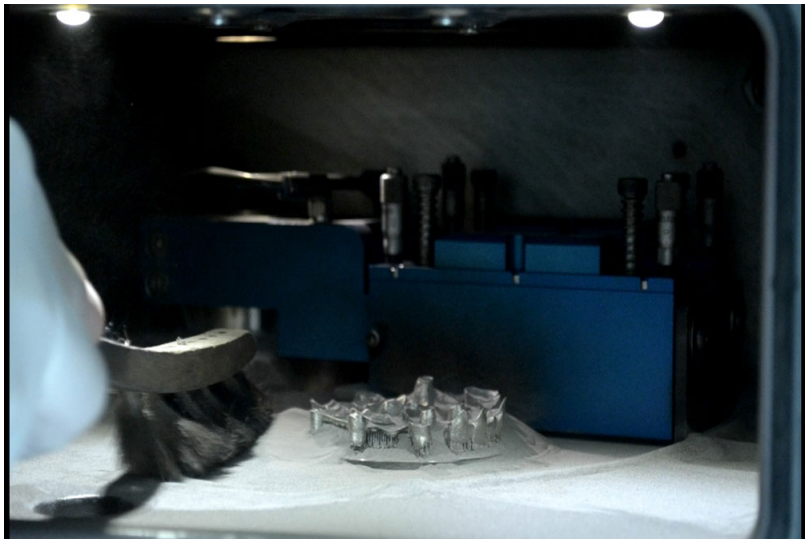

Figure 2: Cleaning in the build chamber with an anti-static brush.

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Other Maintenance Considerations

Performing routine maintenance tasks such as cleaning and inspection includes a risk of inadvertently harming a clean surface. For example, it's important to keep boxes of disposable cleaning tissues or wipes closed during use to avoid accidentally transmitting dust and debris particles. To avoid transferring dust in tight spaces on an L-PBF machine, fold the cleaning wipe several times in squares and use a tweezer to hold it. When dust appears on one part of the square, carefully unfold it and use a clean section of the square. Cleaning swabs can also accidentally spread contaminants when overused, so inspect and replace cleaning swabs frequently.

L-PBF operators should regularly inspect the laser system's protective cover and window for damage, cracks, or burning. Typically, an operator can perform normal maintenance tasks on a laser system without fear of exposure, provided that the laser system remains behind the protective cover and window. Dangers from the laser beam can include operator eye injuries and skin burns. Cleaning the protective window may greatly improve the life of the lens. However, an operator can accidentally smudge, blemish, or transfer oils and particles with dirty gloves, which can damage clean window surfaces and affect machine performance. Operators should change to fresh gloves in order to avoid any potential contamination.

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Lesson: 11/11

L-PBF Maintenance Review

Maintenance Tasks When Machine is On

- Check important safety alerts
- Check atmosphere levels
- Update cybersecurity

Maintenance Tasks When Machine is Off

- Clean lens protective window
- Inspect build chamber liner
- Inspect build chamber window
- Check safety labels

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Class Vocabulary

Access Levels The permissions or restrictions that a user has within a machine or software system. Access levels allow different personnel to use different software functions based on their job role or training.

Additive Manufacturing AM. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. Additive manufacturing methods typically build up layers of material to create an object.

Anti-Static Material that resists the flow of electricity. Antistatic material helps prevent operators from building up static charge or serving as a conduit for electricity.

Argon A colorless, odorless non-reactive gas. Argon is sometimes used with laser powder bed fusion and typically glows blue.

Atmosphere Levels The temperature and humidity inside the build chamber. Atmosphere levels in L-PBF machines are often controlled by using inert gases such as argon and neon.

Build Chamber The area contained within an additive manufacturing (AM) machine where the part is actually built. Many build chambers are enclosed to improve the safety of the manufacturing operation.

Build Controls Software that provides detailed instructions to an additive manufacturing machine. Build controls can included information like exact shape and size of each layer or the temperature of the build chamber.

Build Platform The flat surface on which a part is additively manufactured. The build platform can either be a permanent machine surface from which parts are removed or a surface that can be removed from a machine once the build is complete.

Calibration The comparison and adjustment of a device with unknown accuracy to a device with a known, accurate standard. Calibration reduces variation in L-PBF manufactured part.

Cloud A network of remote servers that can be accessed through the internet. The cloud stores many software applications and can be used to back up data.

Configuration The way in which operating systems are arranged in a machine. Configuration by a technician often affects the way in which an additive machine produces a part.

Cyber Attacks An effort to disrupt, disable, or gain illegal access to a digital device or network. Cyber attacks include hacking, phishing, and installing malware.

Cyber Resiliency The ability to quickly respond to and recover from cyberattacks. Cyber resiliency differs from cybersecurity in that it reacts to breaches that do occur rather than preventing them from happening.

Cybersecurity Protection against criminal or unauthorized access to computer networks, programs, and data. Cybersecurity has become a major industrial concern as networking and connectivity have increased.

Emergency Stop E-stop. A button or switch that brings a machine to a safe, rapid stop. Emergency stops are generally large red buttons against a yellow background.

Feeding System The mechanism by which material enters the point of operation on a machine. The feeding system for laser powder bed fusion uses a recoater or powder blade to spread metal powder evenly across the build platform.

Hopper A container that holds raw materials. Hoppers store powder in an L-PBF machine.

Inert Not chemically reactive. Inert gases provide an atmosphere that does not support chemical reactions that could cause corrosion or fire.

Inspection The examination of a part during or after its creation to confirm that it adheres to quality specifications. During inspection, defects may be identified and corrected.

Laser A device that produces an intense beam of light that can be precisely aimed and controlled. Lasers are used in laser powder bed fusion (L-PBF) and other additive manufacturing (AM) processes.

Laser Powder Bed Fusion L-PBF. An additive manufacturing (AM) method that uses a laser to fuse or melt together layers of powdered polymer, metal, ceramic, or other material. Laser powder bed fusion processes include selective laser sintering (SLS) and selective laser melting (SLM).

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Lens A piece of glass shaped with two opposite curved surfaces that is used for directing and focusing light. A lens can be different sizes for different resolutions.

Local Area Network LAN. A type of wired network that enables communication between a group of computing devices in a specific geographic location. Local area networks are commonly used by manufacturers and other organizations to store and exchange digital information across a large facility.

L-PBF Laser powder bed fusion. An additive manufacturing (AM) method that uses a laser to fuse or melt together layers of powdered polymer, metal, ceramic, or other material. L-PBF processes include selective laser sintering (SLS) and selective laser melting (SLM).

Nitrogen A colorless, odorless, tasteless gas that naturally exists in the atmosphere and is non-reactive. Nitrogen is a common gas used with laser bed powder fusion.

Personal Protective Equipment PPE. An article of clothing or a device worn to minimize exposure to hazards and prevent injury. Personal protective equipment may include safety glasses, safety gloves, ear plugs, respirators, or steel-toed boots.

Planned Maintenance PM. Maintenance that is performed purposely and regularly in order to prevent machine deterioration or breakdown. Planned maintenance is a part of total planned maintenance that keeps machines running efficiently.

Planned Maintenance Optimization PMO. A manufacturing improvement method that increases production and reduces waste through continuous attention to the condition of machines and processes. Planned maintenance optimization's main goal is to maximize equipment usefulness across its lifespan.

Post-Build Basic tasks that an operator carries out after an AM part build is completed. Post-build tasks typically revolve around post-processing functions.

Powder Blade A device that slides forwards and backwards over the surface of a powdered material in a powder bed. Powder blades are used in laser powder bed fusion (L-PBF) systems to smooth powdered material layers over a part or build platform.

Pre-Build Basic tasks that an operator carries out before an AM part build is completed. Pre-build tasks typically revolve around setup functions.

Predictive Maintenance PdM. A maintenance approach that involves collecting data related to machine operation in order to service a machine before maintenance issues arise. A predictive maintenance approach involves performing maintenance on a scheduled basis and helps prevent unscheduled downtime.

Preventative Maintenance A type of maintenance performed while a component is in working order to keep it from breaking down. Preventive maintenance may include lubricating, tightening, and replacing worn parts.

Process Controls A method of checking product performance and batch consistency by regularly measuring and regulating activities that yield a product. Process control involves using software to collect and analyze data with the goal of reducing errors.

Quality An approach to manufacturing that focuses on customer satisfaction. Quality products conform to specifications, are free of defects, and meet the requirements of their anticipated use.

Repeatability The ability of a machine or process to continually deliver consistent and uniform results. Repeatability is crucial for efficiently producing parts that meet quality requirements.

Routers An interconnecting device that can connect two types of networks with different protocols, such as a local area network and a wireless network. A router determines where information packets should go and sends them to their destination by the shortest, most efficient route.

Safety Data Sheet SDS. Mandatory information that must accompany almost every chemical in the workplace except for items like cleaning supplies. A Safety Data Sheet includes details such as the hazards, precautions, and first-aid procedures associated with the chemical.

Servers The physical computer that shares information with other computers within its network. The server performs specialized functions in coordination with other computers.

Sieved Filtered with a device that separates unwanted particulate matter and large particles. Sieves are often used with metal powder for additive manufacturing.

Wide Area Network WAN. A type of network that enables communication between devices across multiple geographic locations. Wide area networks, such as the Internet, are generally open, unsecured networks and are not recommended for exchanging sensitive information.

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